

SETI EAM Optical Link Measurements

[illegible]

SETI EAM Optical Link Data Sheet

EAM S/N# NEC 0238Q003

Board # 006

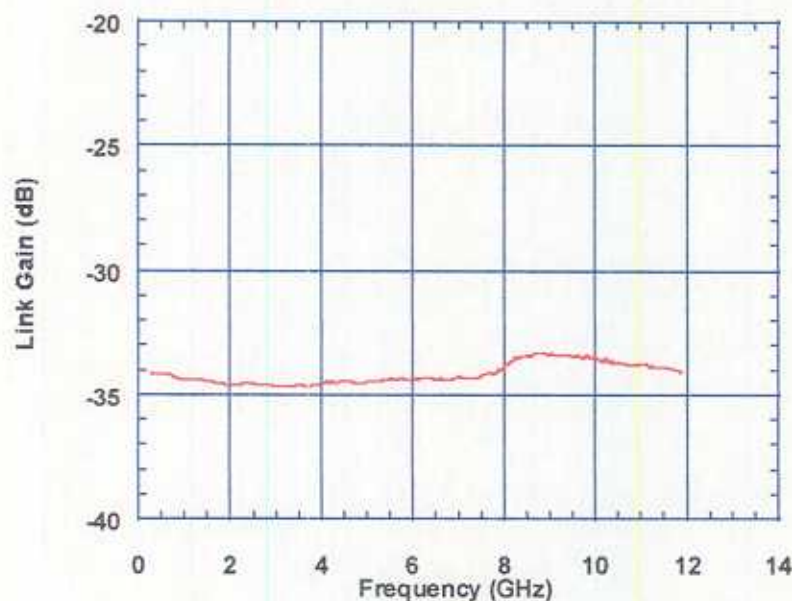
Detector S/N# Discovery 500377

Base plate # EAM006

Test Date 30-Dec 2002

Tested by HVR

Parameter	Conditions	Value	Units
V_{opt} (U3 Pin4)	@ I_{op}	.566	Volts
V_{cc} (U3 Pin6)	For I_{max}	2.867	Volts
V_{mod}	3F @ min	-.657	Volts
3F-F	5GHz@ -10dBm	73.95	dBc
I_{det}	@ I_{opt} $V_{det}=5V$	.907	mA
I_{det} dark current	@ $I_{laser}=0$ mA	.0096	μA
λ	@ I_{op}	1549.3	nm
I_{op}	$R_{th}=10K$	60	mA



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LASER DIODE TEST DATA

TYPE No. NX8560LJ-BC189

CLASS

K

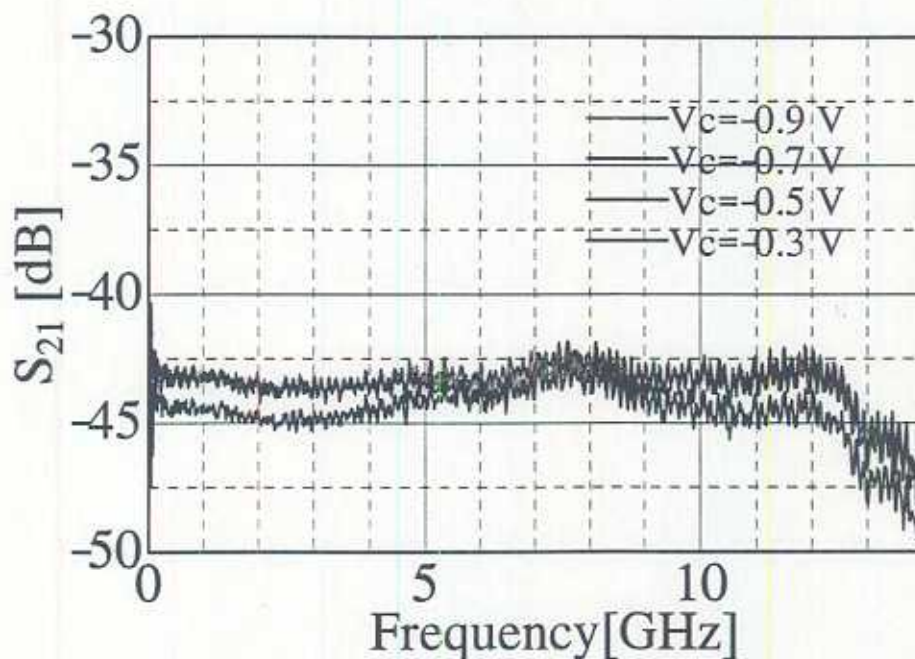
Lot No. 0238Q

S/N 003

Inspected by

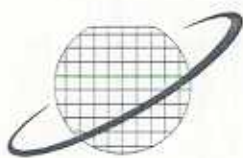


Parameter	Conditions	Value	Unit
I _{op}	-	60.0	mA
V _{mod}	-	2.0	V _{pp}
V _{center}	-	-1.0	V
I _m	@I _{op}	188	μA
T _{set}	-	25.0	°C
R _{th}	T _L =T _{set}	10	kΩ
<P _f >	T _L =T _{set} , On modulation	-1.0	dBm
λ _p	T _L =T _{set} , On modulation	1549.3	nm
RF-ER	T _L =T _{set} , On modulation (without Bessel filter)	11.3	dB
RF-ER	T _L =T _{set} , On modulation (with Bessel filter)	10.2	dB



※ This accompanied data is for an evaluation sample.

Standard accompanied data for NX8560LJ-*C189 is to be discussed.



Discovery Semiconductors, Inc.

119 Silvia Street, Ewing NJ 08628
Tel: (609) 434-1311, Fax: (609) 434-1317

web: www.chipsat.com
email: sales@chipsat.com

186 Princeton-Hightstown Rd, Princeton Jct. NJ 08550
Tel: (609) 275-8667, Fax: (609) 275-0872

An ISO 9001:2000 Certified Company

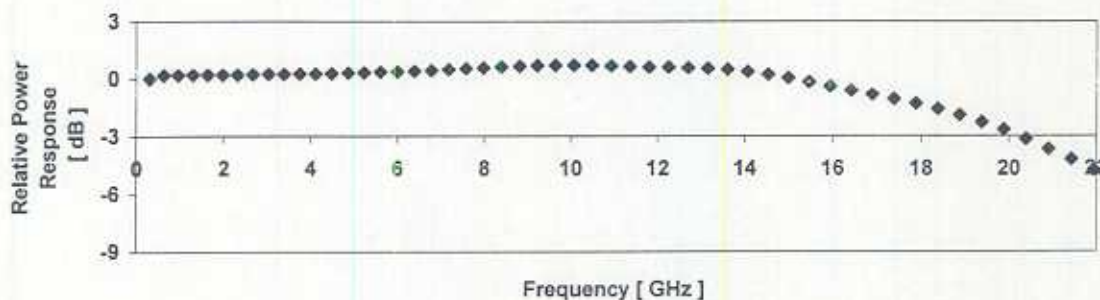
Test Data Report

Customer: Photonic Systems Inc
Part number: DSC50S-39-FC/UPC-K-SP
Description: InGaAs Optical Detector
Tested by: Meilian Qiu

Date: 9/16/02
PO: PSI02-131
WO: 1384
Serial number: 500377
Witnessed by: *[Signature]*

Parameter	Specification	Measured Value	Units
-3dB Bandwidth @ $V_{bd} = +5$ V (Special for PSI)	Min: 14	20.4	GHz
Coupled DC Responsivity @ 1550 nm	Min: 0.70	0.82	A/W
Detector Dark Current @ +5 V	Max: 100	10	nA
Optical Return Loss @ 1550 nm	Min: 30	38	dB

Frequency Response Curve



Remarks:

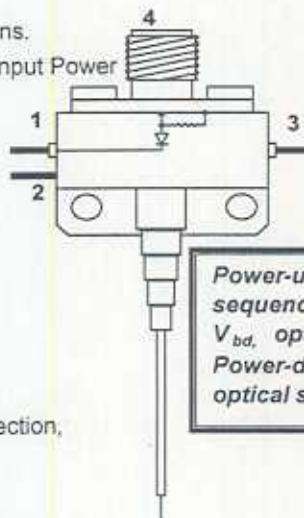
- Please see the package diagram on the right for pin connections.
Recommended $V_{bd} = +3$ V to +7 V depending on the Optical Input Power
- Please see attached data sheet for detail specifications and Maximum Ratings.

Pin Connections:

Pin#	Bias Voltage (V_{bd})
1.	Bias Voltage (V_{bd})
2.	Case Ground *
3.	NC
4.	RF Signal Out

* Observe Polarities

ALWAYS connect ground FIRST, either at case or by RF connection, and ALWAYS disconnect ground LAST.



Power-up sequence:
 V_{bd} , optical signal
Power-down:
optical signal, V_{bd}



Certificate ID: NJ-03-01-02
EAC/SJCNA/CS
Code: 19/3659 3674/33441

DSC50S-SP, REV. B, 08/04/01



RVC Accredited by the RFA

SETI EAM Optical Link Data Sheet

EAM S/N# NEC 0238Q001

Detector S/N# Discovery 500380

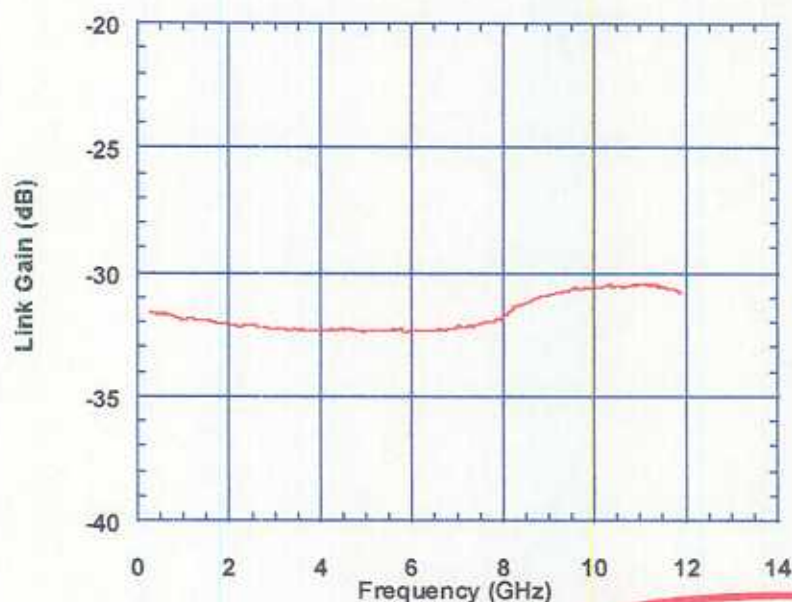
Test Date 30-Dec 2002

Board # 007

Base plate # EAM007

Tested by HVR

Parameter	Conditions	Value	Units
$V_{opt}(U3\ Pin4)$	@ I_{op}	1.052	Volts
$V_{cc}(U3\ Pin6)$	For I_{max}	2.787	Volts
V_{mod}	3F @ min	-.658	Volts
3F-F	5GHz@ -10dBm	75.3	dBc
I_{det}	@ I_{opt} $V_{det}=5V$	1.6278	mA
I_{det} dark current	@ $I_{laser}=0\ mA$	.0123	μA
λ	@ I_{op}	1539.7	nm
I_{op}	$R_{th}=10K$	60	mA



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LASER DIODE TEST DATA

TYPE No. NX8560LJ-BC189

CLASS

K

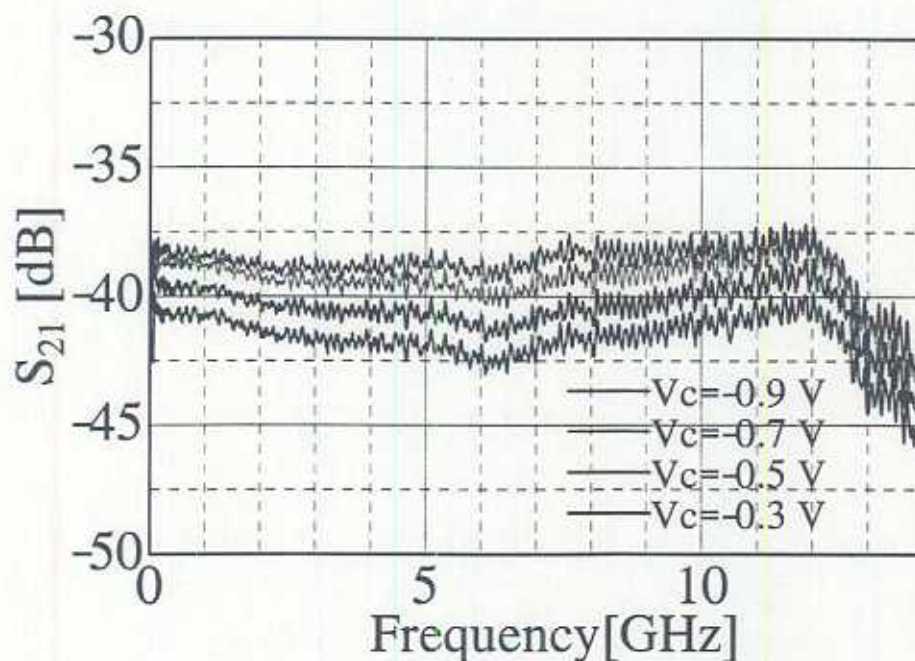
Lot No. 0238Q

S/N 001

Inspected by

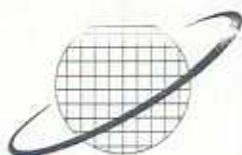


Parameter	Conditions	Value	Unit
I_{op}	-	60.0	mA
V_{mod}	-	2.0	V _{pp}
V_{center}	-	-1.2	V
I_m	@ I_{op}	110	μ A
T_{set}	-	25.0	$^{\circ}$ C
R_{th}	$T_L = T_{set}$	10	k Ω
$\langle PF \rangle$	$T_L = T_{set}$, On modulation	+1.0	dBm
λ_p	$T_L = T_{set}$, On modulation	1539.7	nm
RF-ER	$T_L = T_{set}$, On modulation (without Bessel filter)	11.2	dB
RF-ER	$T_L = T_{set}$, On modulation (with Bessel filter)	10.4	dB



※ This accompanied data is for an evaluation sample.

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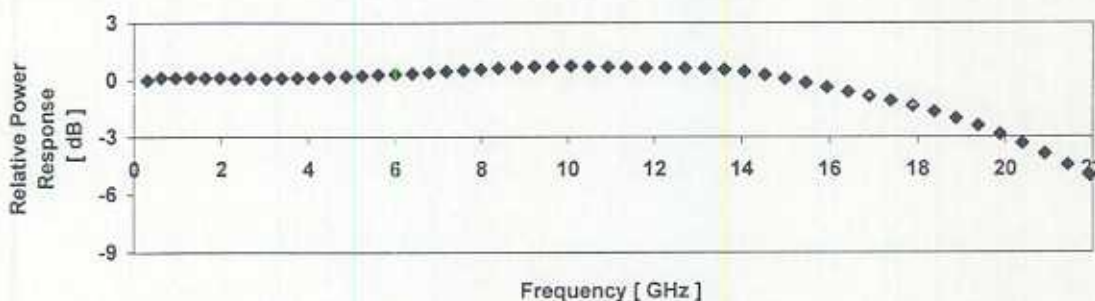
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Description: InGaAs Optical Detector
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Date: 9/16/02
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Serial number: 500380
Witnessed by: *[Signature]*

Parameter	Specification	Measured Value	Units
-3dB Bandwidth @ $V_{bd} = +5V$ (Special for PSI)	Min: 14	20	GHz
Coupled DC Responsivity @ 1550 nm	Min: 0.70	0.78	A/W
Detector Dark Current @ +5 V	Max: 100	11	nA
Optical Return Loss @ 1550 nm	Min: 30	32	dB

Frequency Response Curve



Remarks:

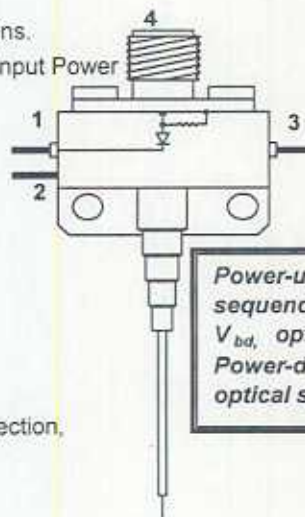
- Please see the package diagram on the right for pin connections.
Recommended $V_{bd} = +3V$ to $+7V$ depending on the Optical Input Power
- Please see attached data sheet for detail specifications and Maximum Ratings.

Pin Connections:

Pin#	1.	Bias Voltage (V_{bd})
	2.	Case Ground *
	3.	NC
	4.	RF Signal Out

* Observe Polarities

ALWAYS connect ground FIRST, either at case or by RF connection, and ALWAYS disconnect ground LAST.



Power-up sequence:
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Power-down:
optical signal, V_{bd}



Certificate ID: NJ-03-01-02
EAC05J01NAUCS
Code: 19/0609 2674/33441

DSC50S-SP, REV. B, 08/04/01



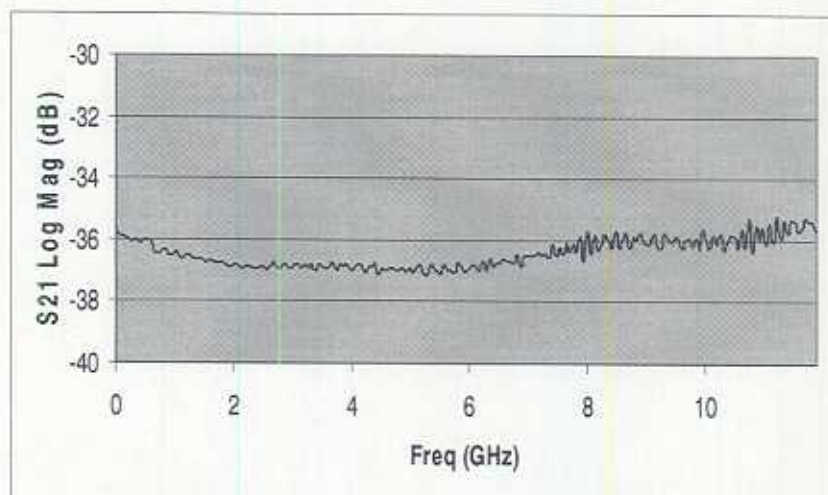
RVC Accredited by the RVC

SETI EAM Optical Link Data Sheet

EAM S/N# NEC 003
 Detector S/N# Discovery 500375
 Test Date 14-Dec 2003

Board # 101
 Base plate # EAM101
 Tested by JD

Parameter	Conditions	Value	Units
V_{opt} (U4 Pin7)	@ I_{op}	1.400	Volts
V_{cc} (U4 Pin7)	For I_{max}	1.640	Volts
V_{mod}	3F @ min	-.278	Volts
3 rd order Distortion	5GHz@ -10dBm	-60.2	dBc
I_{det}	@ I_{opt} $V_{det}=5V$	1.52	mA
I_{det} dark current	@ $I_{laser}=0$ mA	.015	μA
λ	@ I_{op}	1556.3	nm
I_{op}	$R_{th}=10K$	70	mA

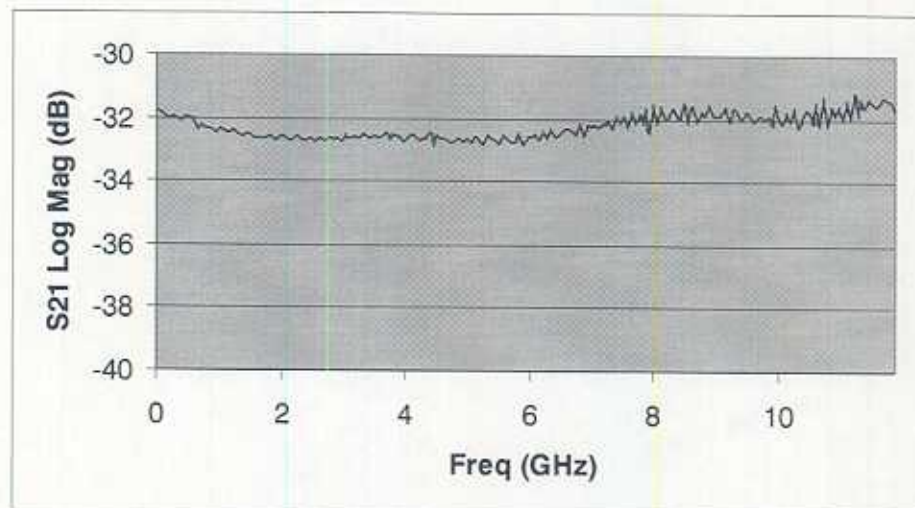


SETI EAM Optical Link Data Sheet

EAM S/N# NEC 014
 Detector S/N# Discovery 500375
 Test Date 15-Dec 2003

Board # 102
 Base plate # EAM102
 Tested by JD

Parameter	Conditions	Value	Units
V_{opt} (U4 Pin7)	@ I_{op}	1.400	Volts
V_{cc} (U4 Pin7)	For I_{max}	1.640	Volts
V_{mod}	3F @ min	-.763	Volts
3 rd order Distortion	5GHz@ -10dBm	-55.85	dBc
I_{det}	@ I_{opt} $V_{det}=5V$	1.10	mA
I_{det} dark current	@ $I_{laser}=0$ mA	.015	μA
λ	@ I_{op}	1537.1	nm
I_{op}	$R_{th}=10K$	70	mA



SETI EAM Optical Link Data Sheet

EAM S/N# 007 LOT 0248Q

Detector S/N# Discovery 500375

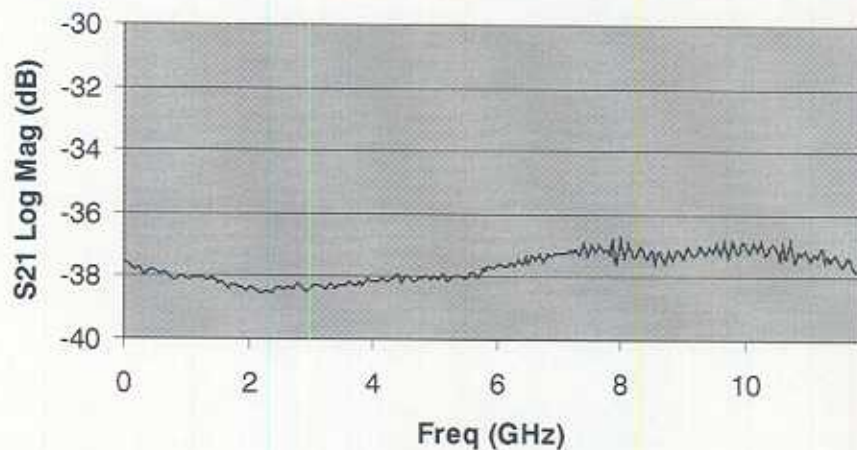
Test Date 31-Dec 2003

Board # 104

Base plate # EAM104

Tested by JD

Parameter	Conditions	Value	Units
V_{opt} (U4 Pin7)	@ I_{op}	1.400	Volts
V_{cc} (U4 Pin7)	For I_{max}	1.640	Volts
V_{mod}	3F @ min	-.185	Volts
3 rd order Distortion	5GHz@ -10dBm	-53.55	dBc
I_{det}	@ I_{opt} $V_{det}=5V$	1.35	mA
I_{det} dark current	@ $I_{laser}=0$ mA	.015	μA
λ	@ I_{op}	1556.2	nm
I_{op}	$R_{th}=10K$	70	mA



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SETI EAM Optical Link Data Sheet

EAM S/N# 005 LOT 0248Q

Detector S/N# Discovery 500375

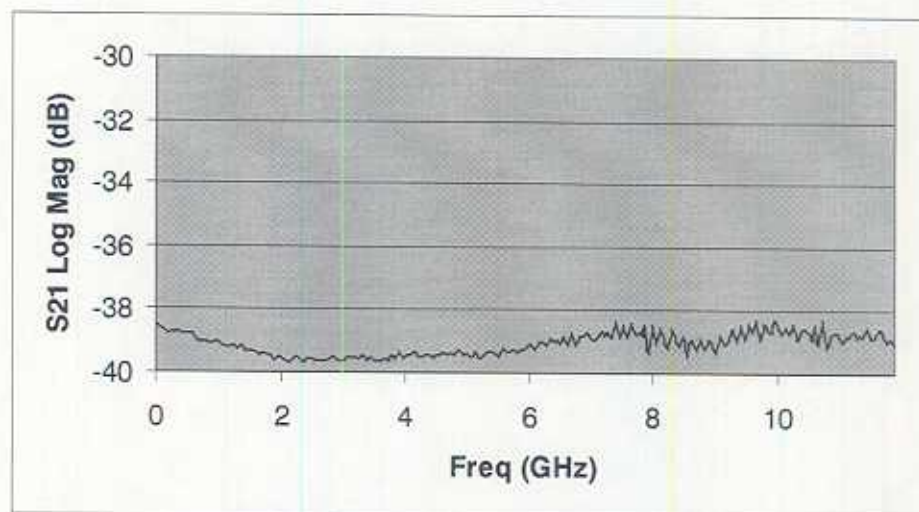
Test Date 31-Dec 2003

Board # 103

Base plate # EAM103

Tested by JD

Parameter	Conditions	Value	Units
V_{opt} (U4 Pin7)	@ I_{op}	1.400	Volts
V_{cc} (U4 Pin7)	For I_{max}	1.640	Volts
V_{mod}	3F @ min	-.263	Volts
3 rd order Distortion	5GHz@ -10dBm	-59.5	dBc
I_{det}	@ I_{opt} $V_{det}=5V$	1.39	mA
I_{det} dark current	@ $I_{laser}=0$ mA	.015	μA
λ	@ I_{op}	1557.0	nm
I_{op}	$R_{th}=10K$	70	mA



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